LOGIC_1V8
LOGIC_3V3
SYS_I2C7_SCL
SYS_I2C7_SDA
CTRL_I2C_SCL
CTRL_I2C_SDA
PWR_MON_ALERT_N
R1001
R1002
R1003
R1004
R1005
U1001
U1002
EMC2305_ADDR
FAN_ALERT_N
C1020
100nF
GND
CPU_FAN_12V
CPU_FAN_PWM
FAN_AUX_12V
MODEM_FAN_12V
MODEM_FAN_PWM
MODEM_FAN_TACH
INTAKE_FAN_12V
INTAKE_FAN_PWM
INTAKE_FAN_TACH
EXHAUST_FAN_12V
EXHAUST_FAN_PWM
EXHAUST_FAN_TACH
F1022
F1023
F1024
NIGHT_LIGHT_SINK
NIGHT_LIGHT_12V
YIS LS102W 12V WARM WHITE
D1070
D1071
NIGHT LED 1
NIGHT LED 2
CPU FAN
MODEM FAN
THA0412AD-TZW3 INTAKE
THA0412AD-TZW3 EXHAUST
C1050
C1051
C1052
TEMP_ALERT_N
TMP117 CM5 / 0x48
TMP117 MODEM / 0x49
TMP117 BOARD POWER / 0x4A
PWR_GREEN_A
LED_PWR_N
PWR_GREEN
D1060
BACKUP_VIOLET_A
LED_BACKUP_N
BACKUP_VIOLET
D1061
WIFI_BLUE_A
LED_WIFI_N
WIFI_BLUE
D1062
CELL_AMBER_A
LED_CELL_N
CELL_AMBER
D1063
TEMP_RED_A
LED_TEMP_N
TEMP_RED
D1064
AUDIO_CYAN
D1065
PWR GREEN
BACKUP VIOLET
WIFI BLUE
CELL AMBER
TEMP RED
AUDIO CYAN
PWR_FLAG
LOGIC_3V3
LOGIC_1V8
FAN_CPU_12V
FAN_AUX_12V
GND

1. 1.8 V TO 3.3 V SYSTEM I2C CONTROL BUS

The control branch is isolated by its own PCA9306. Recheck total bus capacitance with the separate AUDIO-8X8 branch populated.

2. TWO 16-BIT GPIO EXPANDERS

U1001: 7-bit address 0x20; INT is test-point only and firmware polls status.

U1002: 7-bit address 0x21; INT is test-point only and firmware polls status.

TCA9535 outputs power up as inputs. External straps below define safe reset, mute, radio, and amplifier states.

W_DISABLE and WIFI_DISABLE are held active during boot; software releases them only after power and thermal checks pass.

3. EMC2305 FOUR-FAN CONTROL WITH FAILSAFE

ADDR_SEL 4.7k to 3.3 V selects 7-bit SMBus address 0x2E; internal clock is used.

Enclosure channels: 26.00 kHz base / divide 26 = 1.000 kHz PWM; direct duty with tach monitoring, 100% start and 30% minimum running duty.

Fan 1 is panel intake and fan 2 is panel exhaust. CPU and modem fans circulate locally beneath their dedicated mesh/keepout areas.

All four PWM nets pull high when U1020 is absent or unpowered, commanding the selected 4-wire fans to full speed.

6. INDEPENDENT WARM-WHITE PANEL LIGHTS

S1070 pin 5 is the rated low-side load output. Both 12 V LS102W courtesy lights switch together; pin 3 adds a dim green ON ring through R1070.

Hardware-only control: lights remain usable if the CM5 is booting, crashed, or powered down while AUX_12V is available.

4. THREE HIGH-ACCURACY TEMPERATURE SENSORS

The three open-drain TMP117 alerts share TEMP_ALERT_N; firmware reads 0x48/0x49/0x4A to identify the source.

5. SIX TOP-PANEL STATUS LED DRIVERS

TCA9535 sinks LED current. Final LED MPNs and resistor values follow top-panel brightness and color review.

TCA9535 0x20 captures source, fan, temperature, and power-monitor alerts
TMP117 sensors monitor CM5, modem, and board/power zones at 0x48/0x49/0x4A
PWM pull-ups provide full-speed hardware fallback if the controller or firmware fails
EMC2305 controls CPU, modem, intake, and exhaust fans independently
ProComm
Sheet: /
File: Thermal-IO.kicad_sch
Title: Radxa CM5 ProComm Carrier – Thermal and GPIO Control
Size: A0 | Date: 2026-08-14 | **Rev: A1**
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